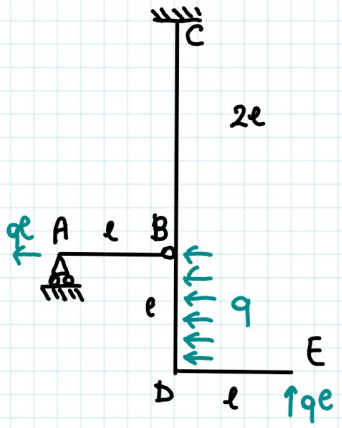
Soluzioni FACSIMILE ESAME

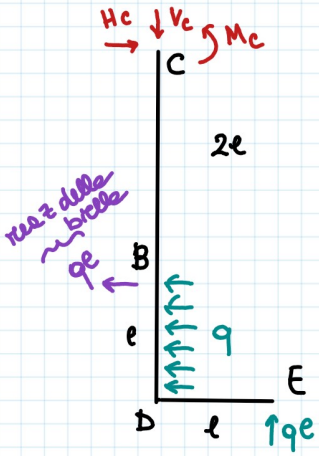
ESERCIZIO 1:



$$\left.\begin{array}{l} gdℓ: 2\times 3 = 6 \\ gdv: 1+2+3 = 6 \end{array}\right\} \text{vincoli ben posti} \rightarrow \text{ISOSTATICA}$$

Noto che: AB è una biella, quindi può solo avere azioni ASSIALI → la reazione del carrello è NULLA.
La biella risulta quindi caricata con una forza assiale (sforzo assiale N) pari a qℓ di TRAZIONE (+).

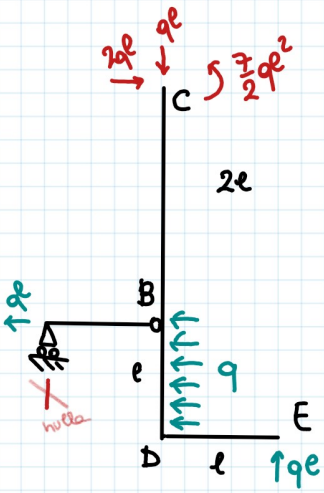
Posso risolvere la struttura CBDE, sostituendo la biella con la sua reazione qℓ.

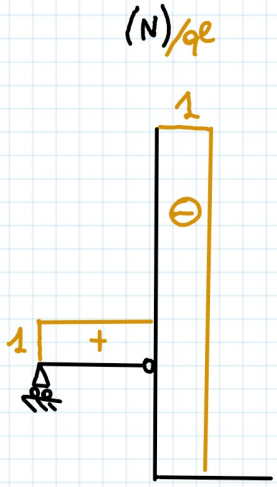


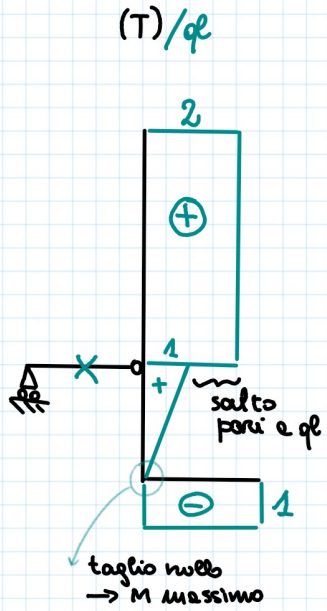
Applico le eq^ni cardinali della statica:

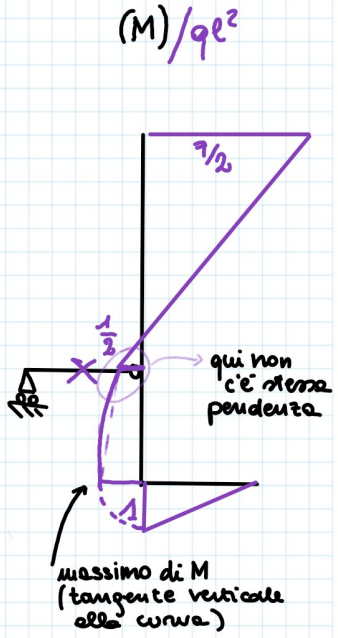
$$\begin{cases} H_C - q\ell - q\ell = 0 & H_C = 2q\ell \\ V_C - q\ell = 0 & V_C = q\ell \\ M(C) = M_C + q\ell\cdot\ell - q\ell\cdot 2\ell - q\ell\cdot\frac{5}{2}\ell = 0 & M_C = \frac{7}{2}q\ell^2 \end{cases}$$

Struttura completa:







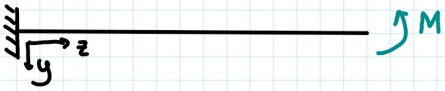


$H_A(\rightarrow) = 0$	$V_A(\uparrow) = 0$	$M_A(\circlearrowleft) = 0$
$H_C(\rightarrow) = 2q\ell$	$V_C(\uparrow) = -q\ell$	$M_C(\circlearrowleft) = \frac{7}{2}q\ell^2$

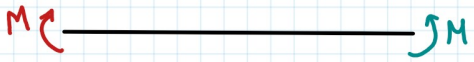
ESERCIZIO 2

Linea elastica $\longrightarrow \varphi_C$

$\searrow \eta_B = \eta\left(z = \frac{\ell}{2}\right)$



Risolvo l'isostatica per calcolare $M(z)$



$M(z) = +M$

$$\left.\right\} \quad \chi = \frac{M(z)}{EI}$$

Trave alla Eulero-Bernoulli:

$$\begin{cases} y'' = -\chi = -\dfrac{M(z)}{EI} \\ y' = -\varphi \end{cases}$$

$$y'' = -\frac{M(z)}{EI} = \frac{1}{EI}(-M)$$

$$y' = \frac{1}{EI}(-Mz) + A$$

$$y = \frac{1}{EI}\left(-M\frac{z^2}{2}\right) + Az + B$$

Mi servono due condizioni a contorno per ricavare A e B

$$\begin{cases} y(z=0) = 0 \\ y'(z=0) = 0 \end{cases}$$ in base ai vincoli

$y(z=0) = 0 \longrightarrow B = 0$

$y'(z=0) = 0 \longrightarrow A = 0$

$$y(z) = -\frac{Mz^2}{2EI} \longrightarrow \eta_B = y\left(z = \frac{\ell}{2}\right) = -\frac{M}{EI}\left(\frac{\ell^2}{8}\right) = -\frac{M\ell^2}{8EI}$$

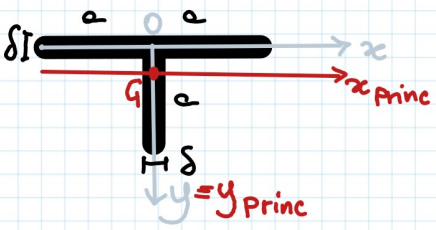
$$y'(z) = -\frac{Mz}{EI} \longrightarrow \varphi = -y' \quad \varphi_C = y'(z=\ell) = -\left(-\frac{M\ell}{EI}\right) = +\frac{M\ell}{EI}$$

ESERCIZIO 3

Sez. sottile aperta con T_y: nascono τ_z

$T_y \longrightarrow \tau_z = \dfrac{T_y S_x^*}{I_x \cdot \delta}$ Formula di Jourawsky, I_x momento d'inerzia della sezione

Verifica sez. con Tresca: $\sigma_{eq} = \sqrt{\sigma_z^2 + 4\tau_z^2}$ $\quad \sigma_z = 0 \rightarrow \sigma_{eq} = 2\tau_z$ Per essere verificata $\sigma_{eq} < \sigma_{amm}$ per $\forall P \in S$



NB la sezione presenta un asse di sym assiale retto → il sistema di riferimento riportato è già CENTRALE.

Devo però traslarlo nel baricentro per far sì che sia PRINCIPALE.

Devo calcolarmi y_G

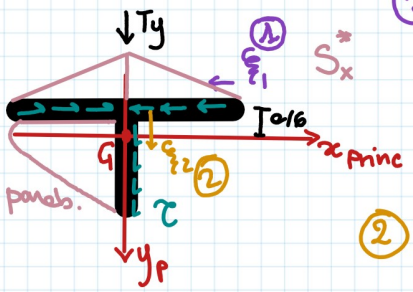
$A = 2a \cdot \delta + a\delta = 3a\delta$

$S_x = a\delta \cdot \dfrac{a}{2} = \dfrac{a^2\delta}{2}$ rispetto a O

$y_G = \dfrac{S_x}{A} = \dfrac{a^2\delta}{2} \cdot \dfrac{1}{3a\delta} = \dfrac{a}{6}$

$G\left(0, +\dfrac{a}{6}\right)$

Traslo l'asse x in G → il sist di rif $x_p G y_p$ è PRINCIPALE.



1 $S_x^{(1)}(\xi_1) = \delta\xi_1 \cdot \left(-\dfrac{a}{6}\right) = -\dfrac{a\delta}{6}\xi_1$ lineare

$\xi_1 = 0 \quad S_x^* = 0$

$\xi_1 = a \quad S_x^* = -\dfrac{a^2\delta}{6}$

$S_x < 0$ τ uscente

La parte di sezione con $x < 0$ è analoga per simmetria

2 $S_x^*(\xi_2) = -\cancel{2}\dfrac{a^2\delta}{\cancel{6}\,3} + \xi_2\delta\left(-\dfrac{a}{6} + \dfrac{\xi_2}{2}\right)$ parabolico

$\xrightarrow{\xi_2 = \frac{a}{6}} S_x^*\left(\dfrac{a}{6}\right) = -\dfrac{a^2\delta}{3} - \dfrac{a}{6}\delta\dfrac{a}{12} = -\dfrac{25}{72}a^2\delta$

$\xrightarrow{\xi_2 = a} S_x^*(a) = -\dfrac{a^2\delta}{3} + \dfrac{1}{3}a^2\delta = 0$

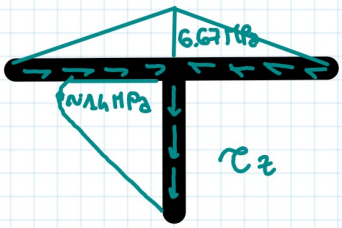
massimo: $\dfrac{\partial S_x^*}{\partial \xi_2} = 0 \quad -\dfrac{a\delta}{6} + \xi_2\delta = 0 \quad \xi_2 = \dfrac{a}{6}$ sul baricentro

questa è una verifica Alle estremità S_x^* è zero

Anche con ξ_2, $S_x^* < 0$, τ uscenti

Poiché $\tau_z \propto a S_x$, l'andamento è lo stesso di S_x^*

$I_x = 2a\delta\left(\dfrac{a}{6}\right)^2 + \dfrac{a^3\delta}{12} + a\delta\left(\dfrac{a}{2} - \dfrac{a}{6}\right)^2 = \dfrac{a^3\delta}{18} + \dfrac{a^3\delta}{12} + \dfrac{a^3\delta}{9} = \dfrac{9a^3\delta}{36} = \dfrac{a^3\delta}{4}$



$S_x^*(\xi_1 = a) = -\dfrac{a^2\delta}{6} \rightarrow \tau_z = \dfrac{T_y \cdot \dfrac{a^2\delta}{\cancel{6}\,3}}{\dfrac{a^3\delta}{\cancel{4}\,2} \cdot \delta} = \dfrac{2T_y}{3a\delta} = \dfrac{2 \cdot 500}{3 \cdot (25) \cdot 2} = 6.67$ MPa

$S_x^*\left(\xi_2 = \dfrac{a}{6}\right) = -\dfrac{25}{72}a^2\delta \rightarrow \tau_z = \dfrac{T_y \cdot \dfrac{25}{\cancel{72}\,48}a^2\cancel{\delta}}{\dfrac{a^3\cancel{\delta}}{\cancel{4}} \cdot \delta} = \dfrac{25}{18}\dfrac{T_y}{a\delta} = \dfrac{25 \cdot 500}{18 \cdot 25 \cdot 2} = 13.89$ MPa

Il punto più sollecitato è il baricentro

Nel baricentro τ_z MASSIMA:

$\sigma_{eq} = 2\tau_z \cong 28$ MPa < 150 MPa (σ_{amm}) Sez. verificata